

Metadata

Title

Bodily Representation of Emotions Across Cultures: A Conceptual Replication of Nummenmaa et al. (2014)

Description

A series of research explored the universality of how people associate different emotions with bodily sensations across 15 countries (Nummenmaa et al., 2014; Volynets et al., 2020). This project is a conceptual replication of experiment 1c from Nummenmaa et al. (2014) on Korean and U.S. samples, where they conducted a comparison of the similarity of bodily representation of emotions across the Finnish and Taiwanese populations.

Contributors

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License

Subjects

Social and Behavioral Sciences

Tags

No data

Study Information

Hypotheses

This project aims to replicate whether the Spearman correlation of the same emotion's representations are more similar across cultures (matched emotion condition; anger-Finnish vs anger-Taiwanese) the correlation of the different emotions' representations across cultures (mismatched emotion condition; anger-Finnish vs happiness-Taiwanese). Consistent with the original study, we also expect to see higher cross-cultural correlations for matched emotions than for mismatched emotions across Korean and U.S. samples.

The original publication did not report an effect size for the difference between these two correlation conditions. We therefore plan to estimate this effect size using data from Volynets et al. (2020), once the relevant cultural metadata becomes available with assistance from the original authors.

Design Plan

Study type

Experiment - A researcher randomly assigns treatments to study subjects, this includes field or lab experiments. This is also known as an intervention experiment and includes randomized controlled trials.

Blinding

No blinding is involved in this study.

Is there any additional blinding in this study?

No data

Study design

This study uses a between-subjects design for the cultural background factor. Because participants cannot be randomly assigned to a cultural group, random assignment is not possible. Instead, native Korean speakers and native English speakers will be separately recruited using the same paradigm.

All participants will complete trials for all emotion categories included in the study. Specifically, they will be presented with one of the seven emotion words (six basic emotions or neutral) in randomized order. To measure the embodied representation of a given emotion word, two silhouettes of bodies will be presented, and participants will color the bodily regions whose activity they felt to be increased or decreased during viewing of each stimulus.

An anticipated challenge when administering this experiment on Prolific is accurately identifying participants' cultural backgrounds. In particular, Korean-speaking participants recruited through Prolific may be highly proficient in English or more immersed in Western cultural contexts, which may reduce cultural specificity. To address this, we will administer a language background questionnaire in addition to the body representation task. This will allow us to more accurately determine whether each participant fits the target cultural group.

No files selected

Randomization

Since participants cannot be randomly assigned to a particular cultural group, this study does not randomize participants at the cultural level. However, this study randomizes the item order for the main experiment, which measures the embodied representation of emotion, as each of the words will occur in randomized order.

Sampling Plan**Existing Data**

Registration prior to creation of data

Explanation of existing data

No data

Data collection procedures

Data will be collected online through Prolific, with two attention checks applied during the language background survey. First Language, Primary Language, and Nationality filters will be applied in Prolific for participant screening. Although there are only a small number of eligible Korean participants, recruiting 18 participants is practically feasible.

No files selected

Sample size

Following the sample size in Nummenmaa et al. (2014), 36 participants will be collected, with 18 participants in each cultural group (United States, Korea.) The two separate culture groups will be collected through two separate Prolific studies.

Sample size rationale

No data

Stopping rule

No data

Variables

Manipulated variables

No data

No files selected

Measured variables

The measured variable is the bodily representations of emotions. To obtain the embodied representation of each emotion per participant, participants will color body regions where they feel increased or decreased bodily activity for each emotion word presented. The resulting data, called "Bodily Sensation Maps (BSM)", will be stored in matrices of paint intensity ranging from 0 to 100 in each pixel. As a result, data from both bodies (the left body indicating activation, the right body indicating deactivation) consists of 50,364 pixels. The activation and deactivation data for each emotion will be combined into a single BSM representing both activations and deactivations by subtracting the deactivation BSM data from the activation BSM data.

No files selected

Indices

No data

No files selected

Analysis Plan

Statistical models

The similarity between the two emotion-wise BSMs from two participants will be measured through pixel-wise correlation (Spearman's rank correlation). Zero values in the heatmaps (i.e., regions without paint) will be filled with Gaussian noise to avoid inflated correlations. Then, based on all possible pairs between participants, correlations of the BSMs of mismatched emotions (for example, sadness and anger) and matched emotions (for example, sadness and sadness) will be calculated. Based on these measures, participant pairs will be classified as one of the two conditions (same culture or different cultures), and a t-test on the correlation values across the two conditions will be performed to measure the difference in correlations across mismatched and matched conditions.

No files selected

Transformations

No data

Inference criteria

No data

Data exclusion

Data will be excluded according to attention check failures, trial completion time, and drawing pattern. First, if the data points from participants failing any of the two attention checks will be excluded. Moreover, data points from participants who had more than one trial completed in less than 5 seconds will be excluded. Lastly, by visualizing each participant's responses for each emotion, drawings exhibiting the same response across 70% or more of the conditions (10 emotions out of 14) in both bodies, or exhibiting overlap in deactivation and activation responses in more than one condition.

Missing data

No data

Exploratory analysis

No data

Other**Other**

Nummenmaa, L., Glerean, E., Hari, R., & Hietanen, J. K. (2014). Bodily maps of emotions. *Proceedings of the National Academy of Sciences*, 111(2), 646-651.

Volynets, S., Glerean, E., Hietanen, J. K., Hari, R., & Nummenmaa, L. (2020). Bodily maps of emotions are culturally universal. *Emotion*, 20(7), 1127.